Recommendations for the Analysis of Leading Indicators in Epidemic Surveillance

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Abstract

Background

Leading indicator analyses are frequently used to assess how epidemic surveillance signals relate to one another. Strong leading indicators can drive early, and better decisions in public health systems.

Methods

We conducted a review of leading indicator studies and extracted common pitfalls in their reporting and analysis, which we used to generate recommendations in retrospective surveillance analyses. Achievable recommendations are provided based on the reporting and analysis components of real-time epidemic time series analysis. We provide contextual examples from the literature of previous studies demonstrating good practice and highlighting areas for improvement.

Conclusion

We present a checklist and workflow that emphases real-time constraints. These recommendations include: causal mechanism reporting, sample coverage and bias, reporting delays and data revisions, multiple event measurement, spatio-temporal granularly, smoothing, transformations, time varying relationships and predictive utility. By following these good practices, analysts can improve research on leading indicators to make analyses more directly actionable for public health decision makers. This work builds on the existing methodology literature by giving a broader framework for overall good practices.

Introduction

The effective mitigation of infectious disease spread, and healthcare resource management often relies on partially complete information. Most infections are unobserved, and those cases detected are subject to severity biases, biological and reporting delays. High-quality reliable population-level signals may not be available in real-time, and when available, may be understood too late to make effective policy or operational decisions. Data sources which reliably indicate future infection trends can support decision making, improve mitigations and communicate risk [1]. We refer to a key surveillance data source, such as hospital admissions with a disease as the "signal", and another data source which may give an early indication of the signal as the "indicator" going forward.

Due to the SARS-CoV-2 pandemic, there has been some increases in data for both COVID-19 and other diseases, generating new indicators to investigate. Indicators are often not traditional epidemiological surveillance (e.g. cases, admissions, deaths), and instead include other collection methods such as internet-based [2], app-based [3] and wastewater data [4]. These indicators are often opportunistically captured, or administrative data designed for alternative purposes. While indicators may often be available more rapidly than traditional surveillance, they are subject to different, and often more substantial biases. "Leading indicators", often called "early warning signals" [5] or "predictors", are widely analysed in epidemiology and play an important role in validating surveillance systems [6] [7]. Often statistics demonstrating how much the indicator and signal agree are reported, as well as estimating lead times between the two. These early warning indicators are frequently harnessed to provide forecasts of signals [8] [9]. Understanding indicator/signal relationships is both useful in outbreaks [10], emerging pathogen situations [11] and established disease systems [12].

However, not all leads between indicator and signal are the same. Leads can be:

- An ***association*** lead – where retrospective evaluations find an alignment between indicator and signal.
- A ***predictive*** lead – the indicator improves out-of-sample performance of a forecast model for the signal at some future time point.
- An ***operational*** lead – a predictive lead which accounts for data latency, reporting delays and real-time factors.

The utility of a leading indicator is inherently based on real-time use, providing decision makers an information edge beyond the signal alone. However, thorough evaluation of leading indicator utility is naturally retrospective. The retrospective analysis of leading indicators can often cause overoptimistic reporting of utility, or present results which are detached from real-world use cases. While there is a diverse range of methods for evaluating leading indicators [13], there is not currently a clear set of guidance for practitioners to consider when designing analyses in epidemic contexts. This mismatch between real-world application and the retrospective methods used to evaluate leading indicators is not inevitable, and by following good practices the conclusions intended for public health action can become more reliable. In this work we will outline key recommendations for the reporting and analyses of leading indicators for epidemiological surveillance systems.

Recommendations

We present nine recommendations for researchers to consider when conducting temporal analysis of epidemiological data, particularly leading indicator analyses. These are summarised in Table 1, with further detail provided in each section.

Type	Number	Name	Description
Reporting	1	Reporting of causal mechanisms	Researchers should articulate how the indicator and signal relate mechanistically, and why a leading relationship is expected.
	2	Sample coverage and bias	Researchers should explain how the populations the signal and indicator draw from overlap with each other.
	3	Reporting delays and data revisions	Researchers should clearly state the reporting delays the indicator and signal are subject to in real time, along with the extent of retrospective revisions.
	4	Single or multiple events	Researchers should articulate whether a subject can influence the indicator or signal once or at multiple time points.
Analysis	5	Smoothing and denoising	If smoothing is necessary, researchers should choose a consistent method for avoiding the use of information unavailable in real-time.
	6	Temporal and spatial granularity	Researchers should avoid generating unmeasured data and prioritise the temporal and spatial scale of the signal.
	7	Transformation: logs, rates, scaling, magnitude and direction	Researchers should mathematically transform their data appropriate to the infection process and sampling method of each data, and align the data with the decision-making process.
	8	Time varying relationships	Researchers should present the evaluation over multiple time periods to demonstrate if leading relationships are consistent.
	9	Predictive utility out of sample and baselining	Researchers should demonstrate indicator lead times through predictive modelling of the signal and compare to a baseline.

Table 1. Recommendation checklist to assess operational utility of leading indicator analyses.

Throughout this manuscript we provide supporting visualisations that were created via simulation of plausible epidemic data, with open code available.

Recommendations: Reporting

1. Reporting of causal mechanisms

Researchers should articulate how the indicator and signal relate mechanistically, and why a leading relationship is expected.

Good leading indicators are not useful because of coincidence. Practitioners should understand how the indicator and signal relate, and the common factors between both. While some indicator-signal relationships may be intuitive, explicitly stating and preferably with a diagram, how the indicator-signal directly or indirectly relate allows readers to have a mental model of the processes involved. More complex relationships have more biases and uncertainty involved, which can transparently be interrogated when made explicit. If the analyst does not know the causal link between indicator and signal, then this is a good opportunity to work with those who can investigate.

Furthermore, from this causal relationship there should be a clear articulation of why, mechanistically, the researchers believe there may or may not be a leading relationship between indicator and signal. In Figure 1 we show explicitly why we think that a Google search might lead in time a hospital admission at the individual level, which helps us infer population level trends. The use of a Directed Acyclical Graph (DAG) can help articulate cause and effect hypotheses clearly [14].

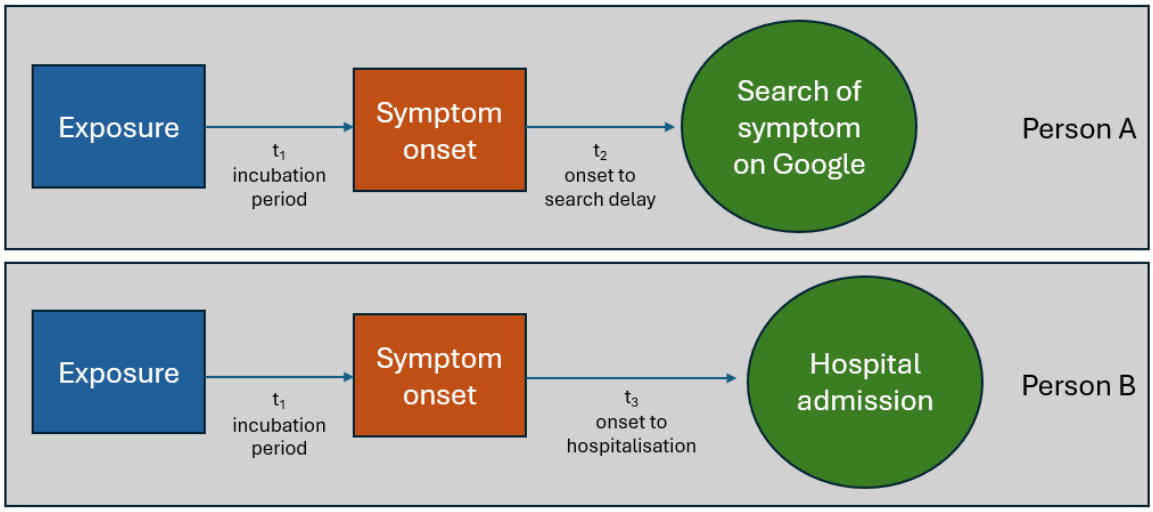

Figure 1. Example Directed Acyclical Graph (DAG) showing the individual level causal chain of events for observations of Google symptom search and hospital admission counts. By showing the different time delays involved at an individual level, we can propose what the leading indicator relationship may be driven by. In this example if $t_2 < t_3$ on average, then in a simple epidemic, we would expect counts of Google symptom searches to lead hospital admission counts by approximately $t_3 - t_2$.

A focus on mechanism can help untangle why lead times are observed, lending credibility to the claims. An example of where this articulation of mechanism strengthens understanding is the work by Dolan et al., investigating the use of non-prescription medications to improve respiratory associated death forecasts [15].

2. Sample coverage and bias

Researchers should explain how the populations the signal and indicator draw from overlap with each other.

Rarely are the signal and indicator data drawn from the same population, which would reduce amount of bias in their comparison. Instead, the indicator may be from a sub-group or different cohort than the signal population. As a result, lead times and utility may be driven by the differences between cohorts, rather than biological or reporting factors. As we expect infection rates to vary in sub-groups of the population, such as by age, sex and location, the surveillance of different sub-groups can generate a leading relationship between indicator and signal.

Furthermore, different sub-groups in a population will have different propensities to engage with measurement for surveillance data. For example, younger individuals may be more likely to engage with online surveillance systems, and those more severely impacted by infection are more likely to engage in healthcare-seeking behaviour. However, in cases where there are unclear relationships between the indicator and signal population, such as non-overlapping geographies, this makes the causal link between the two data more indirect, undermining the leading indicator strength of evidence. Reporting the cohort characteristics of both the indicator and signal sample population, particularly demographic and spatial biases, helps to explain the relationship between surveillance data. Figure 2 shows that there can be clear differences in populations based on the geography covered by different data, and that populations are not guaranteed to overlap between data.

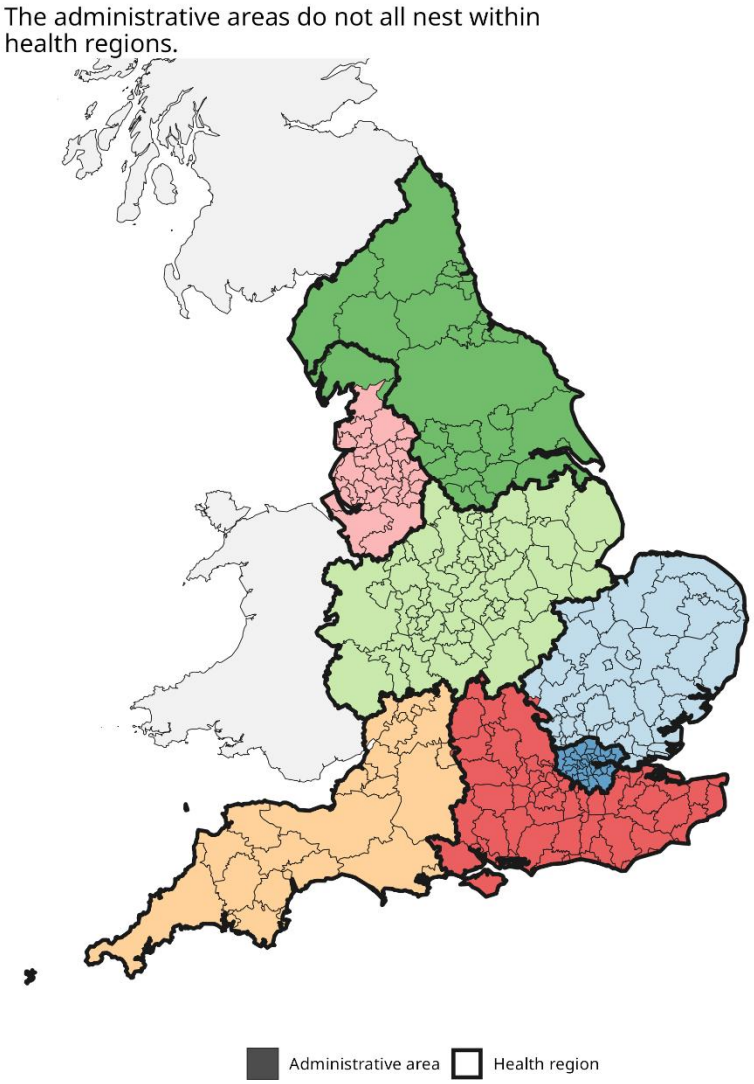

Figure 2. Map of England with healthcare regions (solid black lines) and administrative areas (coloured by the healthcare region their centre resides) highlighted. Data could be gathered at a administrative or a healthcare region, which may not overlap perfectly – highlighting the importance of describing the populations of the indicator and signal.

Not only are individuals from different subgroups more or less likely to engage in surveillance measurement, but differential severity can also impact the timings of infection events. Individuals who are more severely impacted by an infection will seek healthcare earlier than those with mild infections, further complicating the indicator and signal relationship.

Morvan et al. [16] explore wastewater as an indicator of SARS-CoV-2 community prevalence. The authors make clear the difference between the wastewater catchment sites and regions used to estimate prevalence and clearly articulate the mapping process.

3. Reporting delays and data revisions

Researchers should clearly state the reporting delays the indicator and signal are subject to in real time, along with the extent of retrospective revisions.

Real-time surveillance systems must work with the data they receive, however, in retrospective analyses it is often more convenient to use finalised, revised or cleaned versions of data sets. Using finalised data, where reporting delays and backfilling data revisions are not present, biases results and can over-promise utility, producing a "best case" lead time. Practitioners cannot rely on consistent data quality, timeliness or completeness in real time, and therefore latency and the extent of data revisions are critical factors to report.

It is important to understand what information is available at what point in time, as delays and corrections can diminish estimated lead times. An indicator that takes one week to appear in surveillance systems, but has a lead time relative to the signal of two weeks, will have a lead time of one week in practice as some of the lead time is lost to data latency.

Often, real-time individual-level epidemiological data, such as tests reported by collection date, can appear to decline in the most recent version of a time series. This effect is an artifact of reporting delays causing right truncation, rather than a true trend in the time series. This incompleteness can be addressed through nowcasting, which adds uncertainty, or truncating the data until it is complete, adding latency and therefore reducing estimated lead times [17]. Alternatively, the data can be explored according to the date the information was made available for analysis – the report date – which is complete in real-time but induces a bias. The difference between test date and report date is demonstrated via simulation in Figure 3.

In the example of the COVIDcast API from Reinhart et al. [18], a collection of real-time indicators of COVID-19 in the United States, revisions were a key focus of the data collection and this information was readily available for analysis.

Where available, data should be retrospectively analysed in its real-time available version, rather than finalised, to give the most representative estimate of lead times. Data latency for indicator and signal should be clearly reported, alongside the extent of real-time to final data revisions.

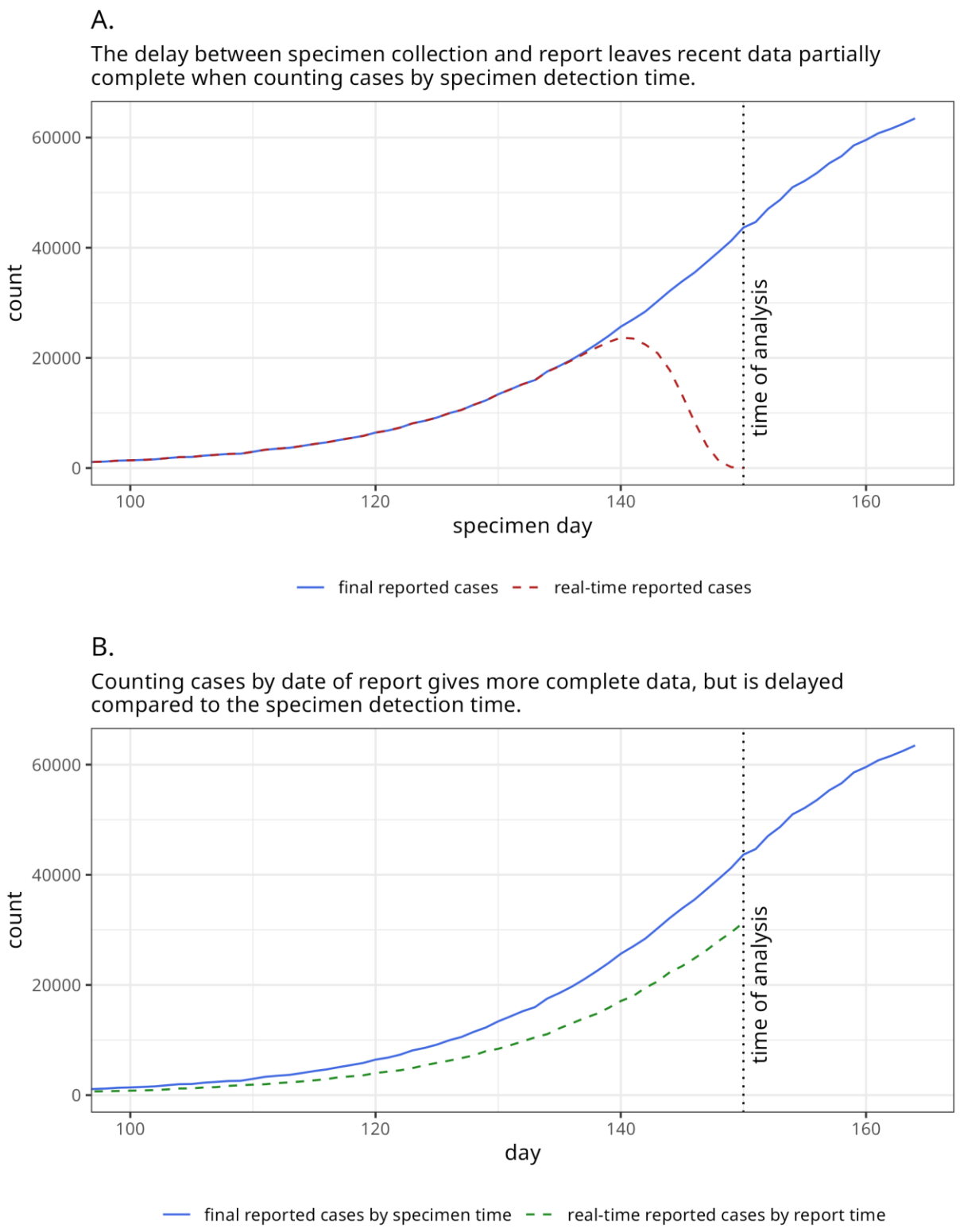

Figure 3. Using a simulation of epidemiological data, the effects of reporting delays are presented, comparing real-time versus final (retrospective) data availability and replicating an analysis from a

linelist. (A) shows that when indexed by a primary (specimen collection) time, then a real-time count will be partially complete due to delays. (B) demonstrates that presenting by the report time helps reduce the incompleteness, but the reporting delay compared to the final data by specimen date impacts the trend.

4. Single or multiple events

Researchers should articulate whether a subject can influence the indicator or signal once or at multiple time points..

Infections and their outcomes can be measured in many ways. At an individual level, infection events are defined to occur at a fixed point in time, such as the onset of symptoms, but other states can be observed for an interval of time, such as being a person being symptomatic. More mechanistically, an individual generally enters a state once at a fixed point, but can stay in that state for multiple time points. These pairs of events and states are common in epidemiology: symptom onset and being symptomatic, hospital admission and being in hospital, infection incidence and prevalence.

In an event-state pair, the state is defined by the event time and duration of individuals in the state. At an individual level, the state is a consequence of the event; in aggregate this causes the appearance of leading relationships between event and state. If the duration of state is trivially short, then the event and state will be matched in time. If the time duration of state is long, then an estimated lead time from event to state will approximate to the mean duration of state. This relationship between event and state time series can be defined by a mathematical convolution. Rather than an indicator and signal; an event-state relationship is more akin to a signal and a transformed version of that same signal – which has limited real-world utility. It is crucial that researchers express whether their indicators and signals are measures of events and/or states. By conflating the two, conclusions about leading indicator relationships can be misleading.

Furthermore, individuals may also appear across multiple data sets which complicates the relationship between indicator and signal. For example, an infected individual could go to a pharmacy which could be detected, and when their symptoms deteriorate they may attend primary or secondary care where again their infection would be recorded.

In the work of D'Auost et al. [19], a range of metrics are compared. A prevalence metric (wastewater, influenced by the duration of viral shedding) is compared with an incidence metric (case count and positivity). Furthermore, these are then compared with a second prevalence measure, hospital beds occupied (influenced by the patient duration of stay). As a result, the leading relationships found will be sensitive to the patient length of stay and duration of viral shedding.

Recommendations: Analysis

5. Smoothing and denoising

If smoothing is necessary, researchers should choose a consistent method for avoiding the use of information unavailable in real-time.

Epidemic time series data are often noisy and challenging to interpret, making smoothing an attractive choice. Smoothing data is used to adjust for statistical artifacts, such as day-of-week effects, mitigate the effect of outliers and to make trends visually clearer. Analysts commonly use methods including rolling averages, splines, and LOESS regression. For example, by using a 7-day rolling average on daily data, the reporting effect of weekends (which may not be relevant to the analysis) can be partially removed, allowing for easier comparison with other data.

However, not all smoothing approaches are appropriate for real-time analysis. Smoothing methods such as splines and LOESS are typically applied to data sets with complete reporting. If we have a signal at time T, $y(T)$, we want to produce a smoothed estimate $\hat{y}(T) = s(y(T))$. Using a smoothing method such as a spline will require data at $y(T + x)$. This is using "future" data to estimate values for today. This is information that we cannot have in real-time, causing overly optimistic estimates for $\hat{y}$ due to "temporal leakage". The difference between what is observed in real-time and in retrospect may be particularly important at crucial epidemic moments, such as at a peak. However, not all smoothing methods rely on future data; a right-aligned rolling average only uses information $y(t = 1), \ldots, y(t = T)$ up to T, which is reproducible in real-time. Though rolling averages (and other methods) can induce a lag to the data, this can be matched between indicator and signal, mitigating the effect. Alternatively, splines using data only up to the most recent data for each time point can be used, though this adds complexity to the analysis. The impact smoothing can have on timing is shown in Figure 4, with different methods producing differently timed epidemic indicators.

Smoothed data can drive stronger correlations between indicator and signals, which may be because of smoothing choices rather than the observed data. By analysing data retrospectively, the analyst can choose tuning methods that will best match the data, causing an overly optimistic leading relationship. In the example of LOESS regression, the analyst can select a span which best approximates how the epidemic data changes over time, but this choice would not be possible without observing the full time series and therefore induces an optimistic bias to the analysis. In some cases it may be ambiguous to the reader whether future information is used in the smoothing process. For example in Kostkova et al. [20], a seven day rolling window is used to smooth social media post counts, but it is not explicitly stated whether this is a centre aligned smooth (using future information) or right aligned (using only past data).

Smoothing that is plausible in real-time, using only information available up to a given time point, can improve the interpretability of leading indicator analyses, but care must be taken to avoid using future information to improve the perceived leading relationship. The unsmoothed data may be more appropriate to report in the analysis, even if the indicator-signal relationship is not as strong, if you believe unsmoothed data would be used by decision makers in real-time. Using smoothed and unsmoothed data in a sensitivity analysis helps demonstrate the robustness of the relationship.

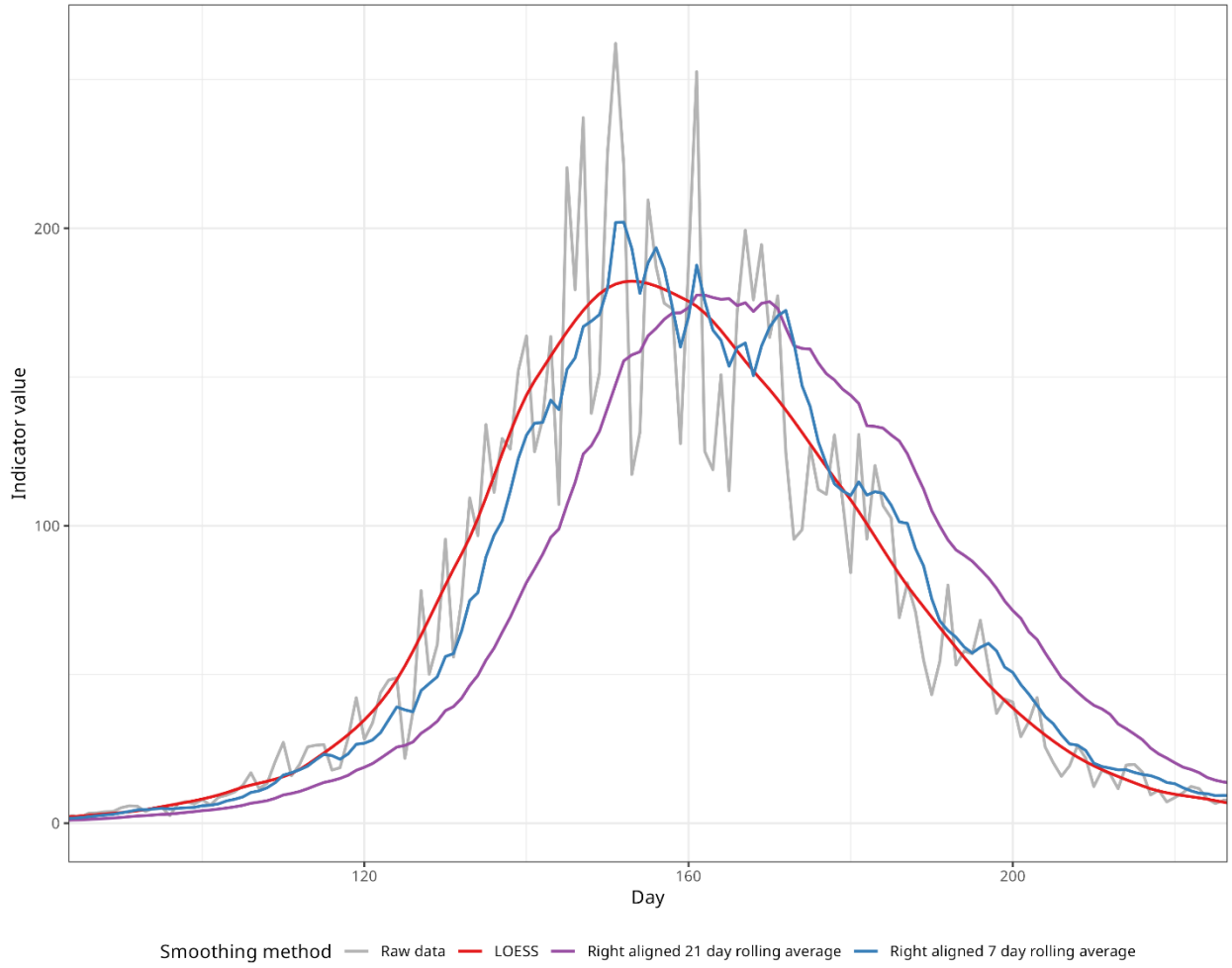

Figure 4. An indicator with a range of smoothing methods applied. Some methods induce a noticeable backward shift in time, such as the 7 and 21 day rolling average. Others may appear closer aligned to the raw data such as the LOESS, but use information that may not be available in real-time.

6. Temporal and spatial granularity

Researchers should avoid generating unmeasured data and prioritise the temporal and spatial scale of the signal.

Non-traditional surveillance data are often not measured at consistent granularities, which makes comparisons challenging. Temporally, we may see daily, weekly, or monthly data, and spatially there may be data at the level of administrative region, county, or nation. Furthermore, some data may not have clearly defined geographical bounds, such as hospital catchments or bespoke areas used only for that data. The preferred temporal and spatial granularity should be that of the signal, rather than the indicator, as it is the data used for decision making.

The mapping from one granularity to another can cause loss in information, for example by converting the indicator from daily to weekly aggregation to align with the signal. Aggregating

granularities is preferred to the alternative, which would require estimating the weekly signal as daily data, in essence estimating data that has not been observed. As with smoothing (Recommendation 6), interpolation can cause unrealistically optimistic associations to be made that may not be knowable in real-time. In cases where one granularity is not strictly a subset of another, such as different administrative geographies, then a common coarser area should be considered. Within Shi et al. [21], exploring dengue forecast modelling, the processing to change the data granularity is clearly stated along with assumptions.

Spatial and temporal granularities are important as they help uncover local and changing heterogeneities in the indicator and signal relationships. This trend heterogeneity in different areas, populations and time periods can strengthen confidence in indicator-signal relationships. However, care must be taken to avoid generating unrealistic data via interpolation and prioritise the granularity of the decision-making signal.

7. Transformation: logs, rates, scaling, magnitude and direction

Researchers should mathematically transform their data appropriate to the infection process and sampling method of each data, and align the data with the decision-making process.

Methods used to evaluate leading relationships may need certain properties from the data, such as linear relationships, stationarity, population normalisation or similar magnitudes, which may require transformation of the data. Furthermore, the signal may be transformed to be more useful for decision makers, such as the growth rate of cases rather than their counts. The indicator and signal will frequently be different types of measurements and may capture different parts of the epidemic process. As a result, transformations on the data should be considered.

Researchers need to engage with decision makers to understand what metric useful in real-time, which will impact the transformations chosen. A common derived metric from surveillance is the epidemic growth rate [22], where the relative change over time is estimated. For methods that rely on assumptions of linearity between indicator and signal, then performing a logarithm of the indicator and signal may be appropriate. The choice of transformation should be driven by prior knowledge or theory, rather than deploying a range of methods to selectively choose one that maximises the leading relationship.

In the work of Rø et al. [23], the researchers transform via statistical modelling a wide range of incidence and prevalence surveillance indicators to the epidemic growth rate, then compare this metric to multiple signals. Furthermore, in McGough et al. [24], a range of transformations including the logarithm and square root are used to compare novel Zika indicators with traditional surveillance metrics. An example of the effect of transformations on analysis is shown in Figure 5.

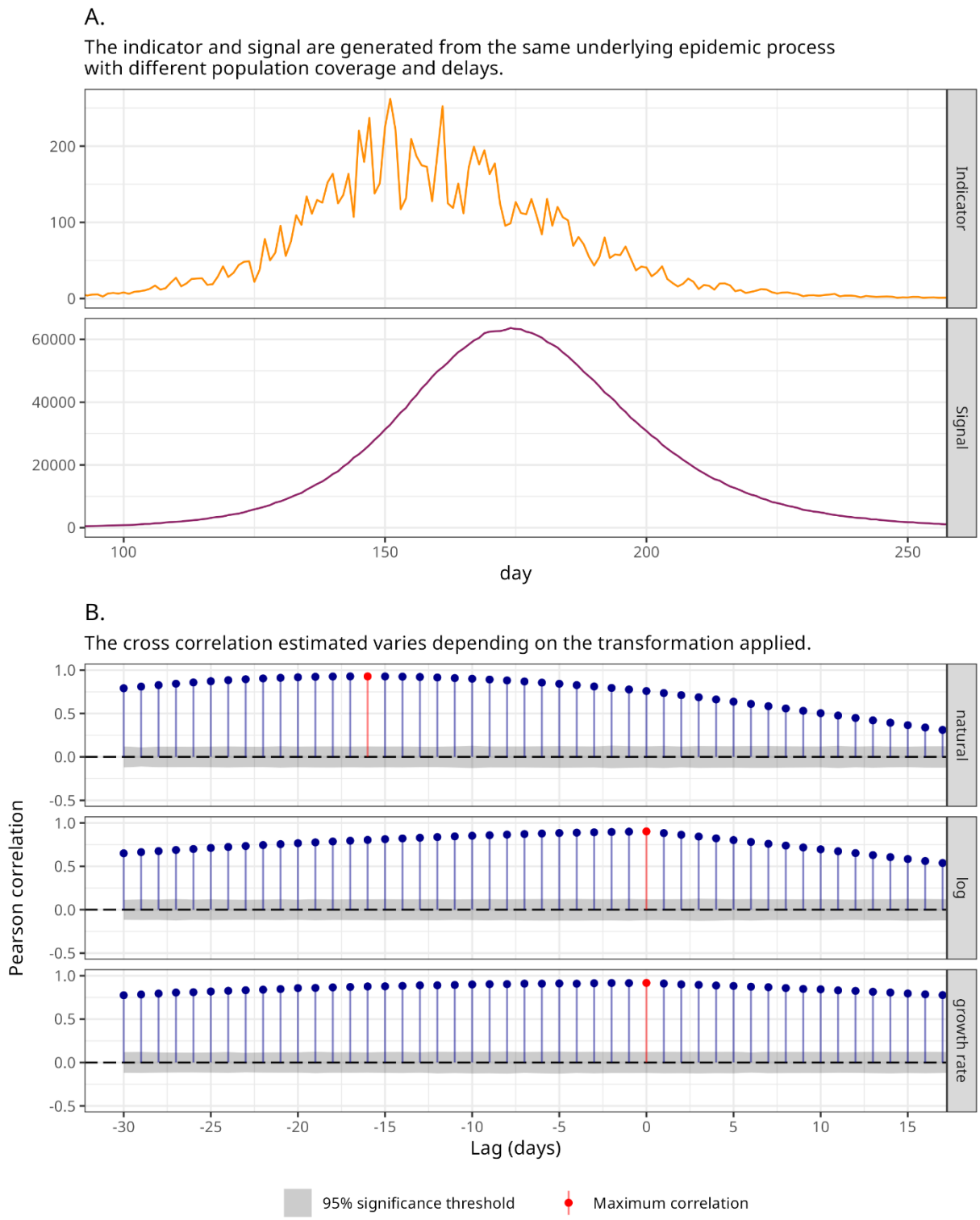

Figure 5. An example leading indicator analysis cross-correlation with bootstrap that is sensitive to scale. (A) shows the indicator and signal over time. (B) demonstrates the cross correlation across scales of the data. There is a clear leading relationship on the natural scale, but this correlation is

lower on a log scale and more similar across a range of lags on the growth rate scale. The "indicator" is a young-age-biased incidence proxy, and the "signal" is cases in the elderly. In this specific example, because there is only one epidemic wave, we have limited information to discern a leading relationship on some scales.

8. Time varying relationships

Researchers should present the evaluation over multiple time periods to demonstrate if leading relationships are consistent.

The relationship between indicator and signal is highly likely to vary over time due to sampling biases (opportunistic sampling), changing biological systems (vaccination strategies / variants / population immunity), control strategies, and other factors. Assuming a fixed relationship over a study period produces an average effect, hiding the changing nature of the relationship. Understanding the underlying mechanism between the leading indicator and signal will help to highlight how rapidly this time-varying relationship is expected to change (Recommendation 1). For example with SARS-CoV-2, there could be changes to variants every few months; alternatively for more seasonal diseases, treating each season separately may be appropriate. Using a single average effect may be convenient for reporting, but hides potentially high utility periods or situations where the leading indicator loses effectiveness.

A simple approach to exploring the time-varying lead effect would be to separate the time period of the study into contiguous time periods, such as 3 month periods, and analyse the relationship separately. If the researcher's mechanistic understanding of the data indicates other periods may be distinct (such as periods of interventions, strain replacement, testing policies) then these could be used. Some methods, such as dynamic time warping [25], incorporate this time-varying effect into their approach explicitly. Alternatively, rolling windows of smaller time steps could be analysed one by one, showing the change in leading relationship over time.

Time varying relationships are important because they can impact decisions made in real time, for example by assuming a constant relationship between signal when in reality the relationship is changing over time one might observe a more optimistic view of an epidemic trend. A rapidly changing relationship between signal and indicator can impact the performance of forecast models, leading to poor prospective model performance.

Care should be taken in identifying the size of time periods to analyse – shorter periods may be under-powered due to less data, while larger periods may average over changing effects. Crucially, all data should be used, rather than selecting only periods of high correspondence. In their analysis of Google Flu Trends with influenza-like-illness surveillance, Cook et al. [26] segment the periods around the 2009 H1N1 pandemic to give a better understanding of how the indicator and signal relate.

9. Predictive utility out of sample and baselining

Researchers should demonstrate indicator lead times through predictive modelling of the signal and compare to a baseline.

While methods that show the correlation between two signals are useful, often the intended purpose of an indicator is to directly predict (through a model or a decision maker's head) the future values of the signal. This is a verifiable statistical problem, whether the inclusion of an indicator in a model improves the predictions of the signal at some future time horizon. Furthermore, a leading relationship of X days, or an arbitrary correlation coefficient is not inherently useful – a baseline signal for analysis should be used to benchmark analysis against.

Epidemic signals are generally autoregressive, due to the onward infection process. As such, autoregressive models which rely on the signal alone should be compared to the same model which also includes the indicator. By doing this the predictive improvement of using the indicator can directly be quantified.

There are numerous examples of leading indicators being assessed using predictive modelling, such as Reich et al. [8] using COVID-19 case reports to improve hospitalisation predictions and McDonald et al. [27] using five different indicators to forecast COVID-19 case signals.

Evaluating a leading indicator by showing improved predictive performance is a strong evidence base for a leading relationship. It is crucial that this is done using out-of-sample evaluation, where models are fit to some training period, and the performance is measured on a separate evaluation set of data to avoid overfitting. The use of proper scoring rules in evaluating forecast performance allows for robust assessment of improvements to the data in an honest manner [28].

Consequences

Each of the nine recommendations can impact the quality of leading indicator analyses. In Table 2 we outline the possible consequences of not considering each recommendation for decision making in real-time.

Type	Number	Name	Consequence if not followed
Reporting	1	Reporting of causal mechanisms	Associations between signal and indicator may be spurious and unrelated to infection dynamics, leading to incorrect poorly timed interventions in real-time.
	2	Sample coverage and bias	Associations identified between signal and indicator may be unknowingly influenced by factors that impact population subgroups differently, which may cause reduced lead times.
	3	Reporting delays and data revisions	Lead times identified retrospectively may be heavily reduced when indicators are used operationally reducing the timeliness of interventions.
	4	Single or multiple events	Lead times identified may be artifacts of measurement rather than due to unique information, reducing the utility of triangulating data for decision makers.
Analysis	5	Smoothing and denoising	Different methods for smoothing may introduce artificial leads which will not

			replicate in real-time, and the use of future information will indicate overly optimistic relationships, reducing utility for decision advantages.
	6	Temporal and spatial granularity	Using indicators in real-time will be practically harder to do without appropriate spatiotemporal mapping, and performance may be lower in practice reducing decision maker confidence in the analysis.
	7	Transformation: logs, rates, scaling, magnitude and direction	Strong lead relationships may be missed without the use of mechanism-informed transformations and relationships identified may not align with how the indicator will be used by decision makers.
	8	Time varying relationships and lags	Decisions may be based on average relationships which may not represent reality causing poorly timed interventions.
	9	Predictive utility out of sample and baselining	Leading relationships may be artifacts of the statistical analyses conducted, rather than practically useful. Without a baseline to compare to, the usefulness of the lead time is not straightforward to assess. Without this decision makers may have false confidence in indicators, reducing decision quality.

Table 2. The consequences of not following each recommendation relating to the operational use of leading indicators.

Discussion

Robust surveillance of infectious disease is a key requirement of good decision-making during epidemics. Understanding how different surveillance systems relate and may predict one another in real time can provide an information edge that improves decisions. However, soundly demonstrating a leading relationship between indicator and signal is challenging retrospectively. There are a range of pitfalls which undermine the credibility of these types of analyses – by accounting for effects discussed in this recommendation list, researchers can improve the robustness of their conclusions for decision makers.

Leading indicator analysis and related work can be valuable for operational use, helping decision makers understand the current state of an epidemic, and improve the performance of important forecasting models. However, poorly-considered leading indicator analyses can be detrimental to model performance and decision making – introducing more confusion rather than clarification. Poor reporting in these analyses makes the research harder to replicate for operational public health teams, and harder to trust the analysis conducted. Without well describing operational challenges, such as latency and revisions, an overly optimistic picture of an indicator is created by researchers, which may be disappointing in real-time. Furthermore, conducting statistical analyses

that do not consider the real-time dynamics and measurement can inflate indicator performance, making the benefits of an indicator challenging to replicate in practice.

We have focused on making recommendations that are generalised to a wide range of surveillance indicators and statistical methods – but specific to real-time challenges. The shape, cadence and challenges associated with new surveillance indicators are myriad and cannot all be discussed, but the principles of the recommendations here should be considered at the start of analyses.

There is substantial prior work on epidemic leading indicator analyses, both exploring indicators themselves and methodologies to identify them [6]. Work by Bloom et al. [13] highlights key caveats with a commonly used approach, the cross-correlation function, which has informed many analyses. Furthermore, the SARS-CoV-2 pandemic has driven improvements in this area [23], as well as highlighted room for improvement in surveillance and analyses [29].

A strong analysis of leading indicator relationships requires knowledge in a range of areas: epidemiology, statistics and real-time data analysis – a research team should cover these areas of expertise. This work will hopefully support improvements in practices for these analyses to avoid common pitfalls to help drive better real-time public health decision making.

Author contributions

Jonathon Mellor: Conceptualisation, Methodology, Investigation, Writing – Original Draft, Writing – Review & Editing, Visualisation, Project administration.
Maria Tang: Methodology, Software, Validation, Writing – Review & Editing, Visualisation.
Robert S. Paton: Conceptualisation, Methodology, Writing – Review & Editing.
Thomas Ward: Conceptualisation, Methodology, Writing – Review & Editing.

Conflict of Interest

The authors declare of no conflicts of interest.

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Data availability

No data was collected as part of this research. Data was generated via simulation, with code available at: https://github.com/jonathonmellor/epidemic-leading-indicator-analysis-recommendations.

Ethics Statement

No ethical approval was required for this methodology focused research using only simulated data.

References

1. Hong, S., et al., *Performance of indicators used in regular risk assessments for COVID-19 in association with contextual factors.* PHRP, 2024. **15**(5): p. 420–428.
2. Milinovich, G.J., et al., *Internet-based surveillance systems for monitoring emerging infectious diseases.* The Lancet Infectious Diseases, 2014. **14**(2): p. 160–168.
3. Quinn, E., et al., *Web-Based Apps for Responding to Acute Infectious Disease Outbreaks in the Community: Systematic Review.* JMIR Public Health Surveill, 2021. **7**(4): p. e24330.
4. Murni, I.K., et al., *Wastewater-based epidemiology surveillance as an early warning system for SARS-CoV-2 in Indonesia.* PLOS ONE, 2024. **19**(7): p. e0307364.
5. Southall, E., et al., *Early warning signals of infectious disease transitions: a review.* Journal of The Royal Society Interface, 2021. **18**(182): p. 20210555.
6. Meckawy, R., et al., *Effectiveness of early warning systems in the detection of infectious diseases outbreaks: a systematic review.* BMC Public Health, 2022. **22**(1): p. 2216.
7. Yang, X., H. Wang, and H. Lu, *Hotspots and Trends in Research on Early Warning of Infectious Diseases: A Bibliometric Analysis Using CiteSpace.* Healthcare (Basel), 2025. **13**(11).
8. Reich, N.G., et al., *Assessing the utility of COVID-19 case reports as a leading indicator for hospitalization forecasting in the United States.* Epidemics, 2023. **45**: p. 100728.
9. Rangarajan, P., S.K. Mody, and M. Marathe, *Forecasting dengue and influenza incidences using a sparse representation of Google trends, electronic health records, and time series data.* PLOS Computational Biology, 2019. **15**(11): p. e1007518.
10. Delecroix, C., et al., *The potential of resilience indicators to anticipate infectious disease outbreaks, a systematic review and guide.* PLOS Global Public Health, 2023. **3**(10): p. e0002253.
11. Yang, K., et al., *Public concerns analysis and early warning of Mpox based on network data platforms-taking Baidu and WeChat as example.* Front Public Health, 2025. **13**: p. 1523408.
12. Mellor, J., et al., *A comparative study of influenza surveillance systems and administrative data in England during the 2022–2023 season.* PLOS Global Public Health, 2024. **4**(9): p. e0003627.
13. Bloom, R.M., D.L. Buckeridge, and K.E. Cheng, *Finding Leading Indicators for Disease Outbreaks: Filtering, Cross-correlation, and Caveats.* Journal of the American Medical Informatics Association, 2007. **14**(1): p. 76–85.
14. Digitale, J.C., J.N. Martin, and M.M. Glymour, *Tutorial on directed acyclic graphs.* J Clin Epidemiol, 2022. **142**: p. 264–267.
15. Dolan, E., et al., *Assessing the value of integrating national longitudinal shopping data into respiratory disease forecasting models.* Nature Communications, 2023. **14**(1): p. 7258.
16. Morvan, M., et al., *An analysis of 45 large-scale wastewater sites in England to estimate SARS-CoV-2 community prevalence.* Nat Commun, 2022. **13**(1): p. 4313.
17. E. Johnson, K., et al., *Baseline nowcasting methods for handling delays in epidemiological data [version 2; peer review: 1 approved, 3 approved with reservations].* Wellcome Open Research, 2026. **10**(614).
18. Reinhart, A., et al., *An open repository of real-time COVID-19 indicators.* Proc Natl Acad Sci U S A, 2021. **118**(51).

19. D'Aoust, P.M., et al., *Catching a resurgence: Increase in SARS-CoV-2 viral RNA identified in wastewater 48 h before COVID-19 clinical tests and 96 h before hospitalizations.* Sci Total Environ, 2021. **770**: p. 145319.
20. Kostkova, P., M. Szomszor, and C.S. Louis, *#swineflu: The Use of Twitter as an Early Warning and Risk Communication Tool in the 2009 Swine Flu Pandemic.* ACM Trans. Manage. Inf. Syst., 2014. **5**(2): p. Article 8.
21. Shi, Y., et al., *Three-Month Real-Time Dengue Forecast Models: An Early Warning System for Outbreak Alerts and Policy Decision Support in Singapore.* Environ Health Perspect, 2016. **124**(9): p. 1369–75.
22. Parag, K.V., R.N. Thompson, and C.A. Donnelly, *Are Epidemic Growth Rates More Informative than Reproduction Numbers?* Journal of the Royal Statistical Society Series A: Statistics in Society, 2022. **185**(Supplement_1): p. S5–S15.
23. Rø, G., et al., *Estimating the trend of COVID-19 in Norway by combining multiple surveillance indicators.* PLOS ONE, 2025. **20**(1): p. e0317105.
24. McGough, S.F., et al., *Forecasting Zika Incidence in the 2016 Latin America Outbreak Combining Traditional Disease Surveillance with Search, Social Media, and News Report Data.* PLOS Neglected Tropical Diseases, 2017. **11**(1): p. e0005295.
25. Berndt, D.J. and J. Clifford, *Using dynamic time warping to find patterns in time series*, in *Proceedings of the 3rd International Conference on Knowledge Discovery and Data Mining*. 1994, AAAI Press: Seattle, WA. p. 359–370.
26. Cook, S., et al., *Assessing Google Flu Trends Performance in the United States during the 2009 Influenza Virus A (H1N1) Pandemic.* PLOS ONE, 2011. **6**(8): p. e23610.
27. McDonald, D.J., et al., *Can auxiliary indicators improve COVID-19 forecasting and hotspot prediction?* Proc Natl Acad Sci U S A, 2021. **118**(51).
28. Bosse, N.I., et al., *Scoring epidemiological forecasts on transformed scales.* PLOS Computational Biology, 2023. **19**(8): p. e1011393.
29. Lipsitch, M., et al., *Infectious disease surveillance needs for the United States: lessons from Covid-19.* Frontiers in Public Health, 2024. **Volume 12 - 2024**.